

Class 9 Science (New NCERT)

Chapter 2: Cell - The Building Block of Life

Introduction

The cell is the basic structural and functional unit of life.

Discovery

Robert Hooke discovered cells in cork (1665). Antonie van Leeuwenhoek observed living cells.

Cell Theory

All organisms are made of cells. Cell is the basic unit of life. New cells arise from pre-existing cells.

Types of Cells

Prokaryotic: No true nucleus. Eukaryotic: True nucleus and membrane-bound organelles.

Cell Organelles

Cell Membrane, Cell Wall, Cytoplasm, Nucleus, Mitochondria, Endoplasmic Reticulum, Ribosomes, Golgi Apparatus, Lysosomes, Vacuoles and Plastids with their functions.

Diffusion: Movement from higher to lower concentration.

Osmosis: Movement of water through a selectively permeable membrane.

Key Points

Mitochondria are the powerhouse of the cell. Ribosomes synthesize proteins.

Chloroplasts are found only in plant cells.